Quantifying Popularity Bias and User-Centric Fairness in Recommender Systems: A Cross-Domain Empirical Study

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***Abstract*—Recommender systems frequently suffer from popularity and exposure biases, leading to a concentration of recommendations on a small subset of mainstream items and disparate accuracy across user groups. This empirical study focuses on two complementary views: *macro-level* exposure inequality (how often items appear in top-K recommendations) and *user-centric* fairness (whether users with niche tastes pay a "niche tax" in RMSE or NDCG@10 relative to mainstream users).**

Six recommendation architectures (collaborative filtering (SVD, ALS, k-NN), content-based filtering (TF-IDF, logistic regression), and a hybrid/ID-based random forest) were evaluated on three public benchmarks (MovieLens 1M, Last.fm 2K, Book-Crossing) and one e-commerce review corpus (Takealot). Matrix factorization models (ALS, SVD) behave as strong popularity amplifiers, with extreme Gini coefficients (e.g., ~ 0.9998 for ALS on Takealot) and very low catalog coverage, while content-based models spread exposure more evenly but often achieve weaker ranking accuracy in sparse settings. User-centric analysis shows a consistent accuracy gap for niche users under neighborhood-based models (k-NN), with dataset- and model-dependent signs for $\Delta\mathrm{RMSE}$. We argue that deployment decisions should audit both aggregate exposure metrics and segment-level accuracy, using ranking-oriented metrics when content models lack calibrated rating heads. A companion manuscript studies simulated feedback loops and retraining dynamics separately.

***Index Terms*—Recommender Systems, Popularity Bias, Collaborative Filtering, Content-Based Filtering, Fairness, User-Centric Evaluation, Empirical Evaluation**

I. Introduction

Personalized recommender systems drive discovery on digital platforms but are trained on interaction data skewed toward popular items. This skew yields *popularity bias* [1]: algorithms over-recommend mainstream items and under-serve the long tail, homogenizing experiences and disproportionately affecting users with niche tastes [2].

Prior work often studies such biases on dense movie-rating benchmarks; such settings can under-represent the extreme sparsity and catalog scale of modern e-commerce. We present a cross-domain study that anchors MovieLens, Last.fm, and Book-Crossing against a large e-commerce corpus (Takealot), and we critique mapping content-similarity scores to explicit rating scales for RMSE, advocating ranking-based metrics (e.g., NDCG@10) where appropriate.

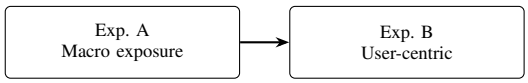


Fig. 1. Empirical pipeline: macro-level top-K bias metrics, then user-centric fairness by niche/diverse/mainstream tertiles.

We address two questions: *(i)* Do algorithms concentrate exposure on already popular items, and how does this differ between collaborative and content-based families? *(ii)* Do users with niche interaction histories suffer a "niche tax" in RMSE or NDCG@10 versus mainstream users?

Outline. Section II situates the work. Section III describes datasets, models, and protocols for macro and user-centric evaluation. Section IV reports results. Section V concludes. Simulated feedback with iterative retraining is analyzed in a companion paper.

II. Related Work

Popularity bias and learning-to-rank have been studied extensively in recommender systems [1]; unfairness arising from popularity skew is discussed in [2]. Standard benchmarks include MovieLens [3], HetRec/Last.fm-style data [4], and Book-Crossing [5]. Implementations often use libraries such as Surprise for k-NN and SVD [6]. Our contribution is a unified empirical comparison across domains, including a sparse retail catalog, with macro exposure metrics and user-centric segment analysis.

III. Preliminaries and Methods

A. Overview

We employ a two-part design: macro-level exposure metrics, then user-centric segment analysis. Fig. 1 sketches the order of experiments (macro recommendations are evaluated for concentration and coverage; the same splits feed segment labels for RMSE and NDCG).

B. Datasets

We use MovieLens 1M (explicit ratings, genre metadata), Last.fm 2K (implicit counts mapped to 1–5 percentiles; tags

TABLE I
CORPORA USED IN THE CROSS-DOMAIN COMPARISON.

Dataset	Domain	Note
ML 1M	Movies	Large movie-rating benchmark; genres.
Last.fm 2K	Music	Counts $\rightarrow$ ratings; tags.
Book-Crossing	Books	Sparse ratings; text.
Takealot	E-commerce	Large sparse catalog.

and artists), Book-Crossing (explicit ratings; author/publisher text; implicit zeros removed), and Takealot (e-commerce reviews; sparse interactions; title, brand, department). A k-core filter (≥ 10 interactions per user and item) stabilizes public graphs. Table I lists corpora.

C. Models

Baselines: user-based k-NN; SVD and ALS (matrix factorization); TF-IDF and per-user logistic regression on TF-IDF metadata; random forest on label-encoded user and item IDs. SVD and k-NN use Surprise [6].

D. Protocols

Macro (top-10): Gini on recommendation frequencies, ARP, catalog coverage, Spearman ρ between training popularity and exposure, plus auxiliary diagnostics (extended panels, log–log and CDF plots) logged in the experiment run for analysis but not all reproduced as figures here.

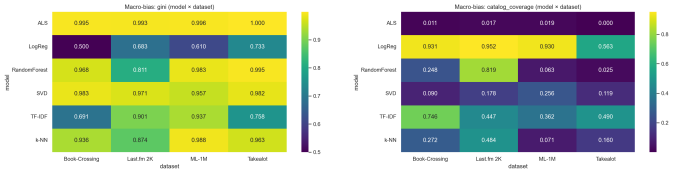
User-centric: Users are split into Niche, Diverse, and Mainstream *tertiles* by the mean popularity (in the training log) of items they interacted with. We report $\Delta\mathrm{RMSE} = \mathrm{RMSE}_{\mathrm{niche}} - \mathrm{RMSE}_{\mathrm{mainstream}}$ and NDCG@10. Content models without native rating scalars use a global training mean for RMSE; segment RMSE can be identical within such models; we emphasize NDCG@10 for cross-family comparison.

IV. RESULTS AND DISCUSSION

A. Macro-level bias

Collaborative models concentrate recommendations on a few items (high Gini), while content-based models such as logistic regression on TF-IDF achieve higher catalog coverage. **Fig. 2** is the main macro summary: each cell is one model–dataset pair. Darker Gini cells indicate stronger inequality in how often items appear in top-10 lists; higher coverage means a larger fraction of the catalog receives at least one recommendation. Reading down the Takealot column, ALS is extreme on both axes (near-maximum Gini, near-zero coverage), whereas LogReg improves coverage substantially at lower Gini than ALS. The three public benchmarks (MovieLens 1M, Last.fm, Book-Crossing) are aggregated consistently; Takealot uses the official split and the same metric code. Table II gives two representative pairs (ALS vs. LogReg) to anchor the heatmap.

SVD shows strong positive ρ between popularity and exposure on ML-1M (≈ 0.52). LogReg reaches high coverage (e.g., $\sim$93% on ML-1M, $\sim$56% on Takealot).



(a) Gini (b) Catalog coverage

Fig. 2. Macro exposure inequality (Gini on recommendation counts) and catalog coverage, by model and dataset. Takealot illustrates e-commerce sparsity: collaborative factorization can collapse to recommending almost no distinct items (low coverage).

TABLE II
REPRESENTATIVE MACRO METRICS (ALS VS. LOGREG).

Dataset	Model	Gini	Cov.
Takealot	ALS	0.9998	0.00029
Takealot	LogReg	0.733	0.563
ML 1M	ALS	0.996	0.019
ML 1M	LogReg	0.610	0.930

B. User-centric fairness

We summarize the niche–mainstream gap with **Fig. 3** only: bar height is $\Delta\mathrm{RMSE} = \mathrm{RMSE}_{\mathrm{niche}} - \mathrm{RMSE}_{\mathrm{mainstream}}$ for each model and dataset. Positive bars mean niche users have higher error than mainstream users under RMSE. k-NN shows a clear positive gap on Book-Crossing in our run; SVD on Last.fm can reverse the sign. Content models that use a global mean for every test prediction (ALS, TF-IDF, LogReg in our implementation) show *flat* $\Delta\mathrm{RMSE}$ across models within a dataset; the bar chart should be read together with NDCG@10 from the tables in the artifact, not RMSE alone. On Takealot, NDCG@10 is uniformly low (< 0.02) for all models, reflecting extreme sparsity; the small gaps in the figure are less informative than ranking metrics there.

C. Synthesis

Popularity bias is model- and domain-dependent. ALS can collapse to extreme inequality on massive sparse catalogs; content methods improve coverage but complicate RMSE comparison without calibrated scores. Systems should be audited for *both* aggregate exposure (Fig. 2) and equitable accuracy across segments (Fig. 3).

V. CONCLUSION

We presented an empirical framework comparing six recommenders across three public datasets and Takealot, measuring macro concentration and niche versus mainstream errors and NDCG@10. Matrix factorization can act as a popularity amplifier; content-based methods spread exposure but require careful ranking-oriented evaluation. Future work includes calibrated rating heads for content models, statistical significance across splits, richer user partitions, and, in a companion manuscript, simulated feedback loops with retraining.

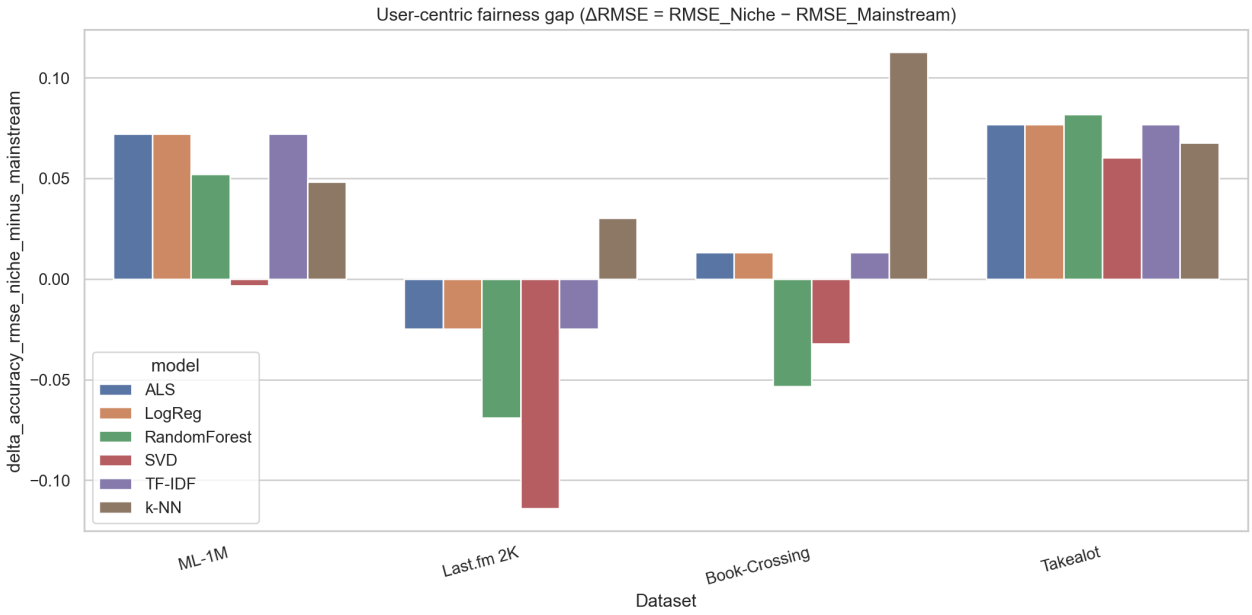


Fig. 3. User-centric fairness gap (ΔRMSE, niche minus mainstream) by dataset and model. Global-mean fallbacks for some content models can equalize segment RMSE; interpret alongside NDCG@10.

Acknowledgment

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